NAME : SHIWANSHU KUMAR JHA
MOBILE_NUMBER : 7979756698
EMAIL_ID : shiwanshukumarjha7@gmail.com
GIT_REPO : https://github.com/shiwanshu97/GIT_Assessment.git

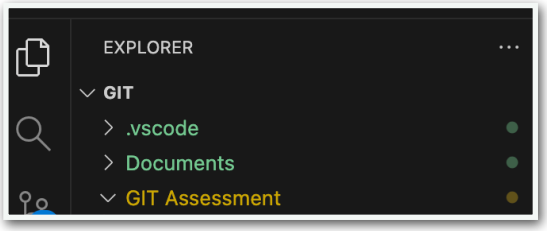
Question 1: Project Initialisation & First Push

Objective : Set up a new Git project and push it to a remote repository.

* Tasks :-

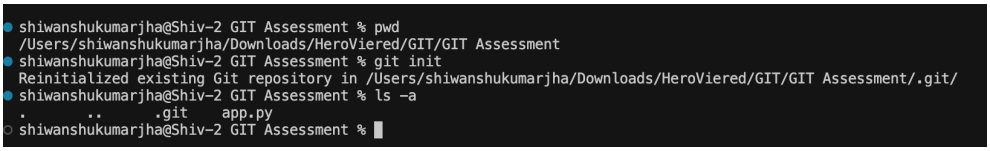
1. Create a new folder for your project

   Git Assessment is a new folder

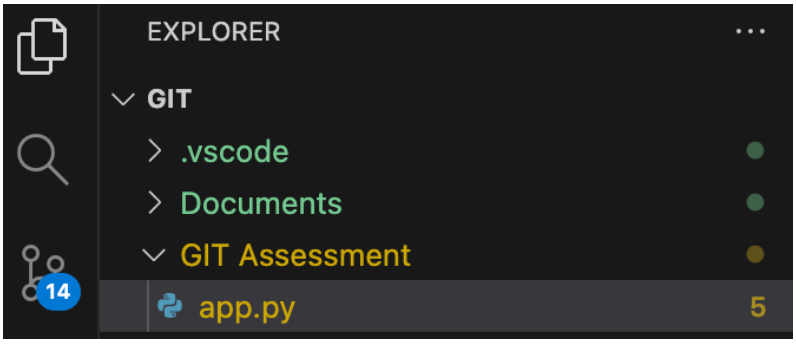


2. Initialise Git repository

   Initialised git repository using git init command



3. Create a file named app.py and add some Python code



```python
app.py 5 ×
GIT Assessment > app.py > add_student
from flask import Flask, render_template, request, redirect, url_for
from flask_pymongo import PyMongo
from bson.objectid import ObjectId
from dotenv import load_dotenv
import certifi
import os

# Load env vars
load_dotenv()

app = Flask(__name__)
app.config["MONGO_URI"] = os.getenv("MONGO_URI")
app.secret_key = os.getenv("SECRET_KEY")

# Use certifi CA bundle explicitly for cross-platform TLS reliability
# (notably fixes common macOS certificate verification failures).
mongo = PyMongo(app, tlsCAFile=certifi.where())
```

4. Check the current Git status

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.py

nothing added to commit but untracked files present (use "git add" to track)
```

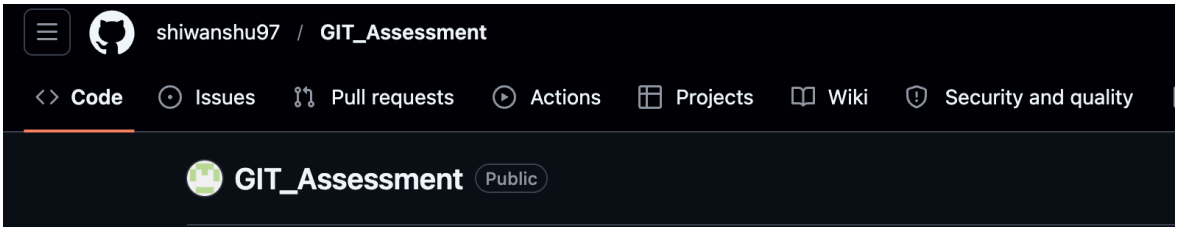
5. Stage the file

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git add app.py
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   app.py

shiwanshukumarjha@Shiv-2 GIT Assessment %
```

6. Commit with a meaningful message

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git commit -m "Initialize Git project and add app.py with initial Python code"
[master (root-commit) 367e75f] Initialize Git project and add app.py with initial Python code
 1 file changed, 63 insertions(+)
 create mode 100644 app.py
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
nothing to commit, working tree clean
```

7. Create a remote repository (GitHub or similar)



8. Add the remote (origin) to your local repo

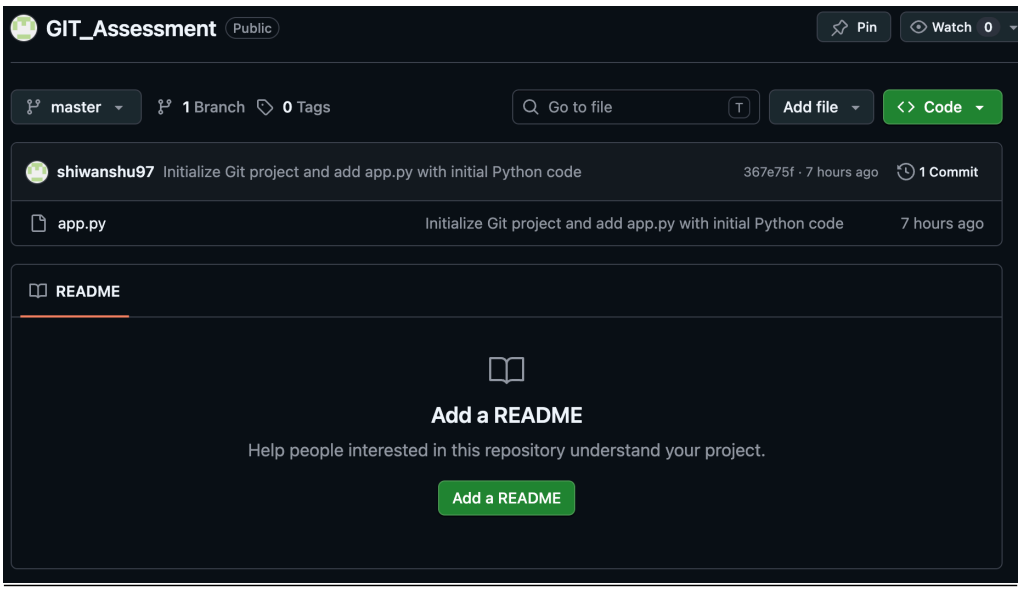
```
shiwanshukumarjha@Shiv-2 GIT Assessment % git remote add origin https://github.com/shiwanshu97/GIT_Assessment.git
shiwanshukumarjha@Shiv-2 GIT Assessment % git remote -v
origin  https://github.com/shiwanshu97/GIT_Assessment.git (fetch)
origin  https://github.com/shiwanshu97/GIT_Assessment.git (push)
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

9. Verify the remote configuration

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git remote -v
origin  https://github.com/shiwanshu97/GIT_Assessment.git (fetch)
origin  https://github.com/shiwanshu97/GIT_Assessment.git (push)
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

10. Push your code to the remote repository

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
* master
shiwanshukumarjha@Shiv-2 GIT Assessment % git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 986 bytes | 986.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shiwanshu97/GIT_Assessment.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
shiwanshukumarjha@Shiv-2 GIT Assessment %
```



Question 2: Working with Changes & History

Objective : Track code changes and manage commit history properly.

* Tasks :-

1. Modify app.py by adding new functionality.

```
# About page
@app.route('/about')
def about():
    return "Student Management System using Flask and MongoDB"
```

2. Check what changes are made before staging.

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

3. View differences in the file.

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git diff
diff --git a/app.py b/app.py
index 0f66ac6..ef4927e 100644
--- a/app.py
+++ b/app.py
@@ -59,5 +59,10 @@ def delete_student(student_id):
     mongo.db.students.delete_one({"_id": ObjectId(student_id)})
     return redirect(url_for('index'))

+# About page
+@app.route('/about')
+def about():
+    return "Student Management System using Flask and MongoDB"
+
 if __name__ == '__main__':
     app.run(host="0.0.0.0", debug=True, port=5001)
\ No newline at end of file
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

4. Stage only specific changes (if possible)

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git add ./app.py
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app.py
```

5. Commit with a clear message.

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git commit -m "Add about page route to Flask application"
[master 79578d6] Add about page route to Flask application
 1 file changed, 5 insertions(+)
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

6. Make another change in app.py

```
# Health check route
@app.route('/health')
def health():
    return "Application is running successfully"
```

7. Stage all changes

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git add .
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app.py

shiwanshukumarjha@Shiv-2 GIT Assessment %
```

8. Commit again

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git commit -m "Add health check route to Flask application"
[master 5a4ae24] Add health check route to Flask application
 1 file changed, 5 insertions(+)
```

9. View full commit history

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git log
commit 5a4ae2420dbb151ccf8f0ff175f14d0a597c9ed0 (HEAD -> master)
Author: shiwanshu97 <shiwanshukumarjha7@gmail.com>
Date:   Thu May 7 11:37:41 2026 +0530

    Add health check route to Flask application

commit 79578d63e09f84f92666a6c3240d8179ca024c87
Author: shiwanshu97 <shiwanshukumarjha7@gmail.com>
Date:   Thu May 7 11:24:09 2026 +0530

    Add about page route to Flask application

commit 367e75f1f9f2d79a8ec9710625b41fe9fea2befb (origin/master)
Author: shiwanshu97 <shiwanshukumarjha7@gmail.com>
Date:   Wed May 6 15:32:30 2026 +0530

    Initialize Git project and add app.py with initial Python code
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

10. View compact (one-line) history

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git log --oneline
5a4ae24 (HEAD -> master) Add health check route to Flask application
79578d6 Add about page route to Flask application
367e75f (origin/master) Initialize Git project and add app.py with initial Python code
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

Question 3: Branching & Feature Development

Objective : You are developing a new feature separately to avoid affecting the main code.

* Tasks :-

1. Create a new branch (e.g., feature-update)

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
* master
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch feature-update
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
  feature-update
* master
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

2. Switch to the new branch

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git checkout feature-update
Switched to branch 'feature-update'
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
* feature-update
  master
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

3. Modify app.py with new feature logic

```
# Contact page route
@app.route('/contact')
def contact():
    return "Contact us at support@studentapp.com"
```

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch feature-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

4. Stage and commit the changes

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git add app.py
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch feature-update
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app.py

shiwanshukumarjha@Shiv-2 GIT Assessment % git commit -m "Add contact page route in feature branch"
[feature-update 639f191] Add contact page route in feature branch
 1 file changed, 5 insertions(+)
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

5. Switch back to the main branch

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
* feature-update
  master
shiwanshukumarjha@Shiv-2 GIT Assessment % git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
  feature-update
* master
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

6. Merge the feature branch into main

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git merge feature-update
Updating 5a4ae24..639f191
Fast-forward
 app.py | 5 +++++
 1 file changed, 5 insertions(+)
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

7. Verify changes are merged

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git log --oneline
639f191 (HEAD -> master, feature-update) Add contact page route in feature branch
5a4ae24 Add health check route to Flask application
79578d6 Add about page route to Flask application
367e75f (origin/master) Initialize Git project and add app.py with initial Python code
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

8. Delete the branch safely

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
  feature-update
* master
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch -d feature-update
Deleted branch feature-update (was 639f191).
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
* master
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

9. Try force deleting a branch (create a dummy branch for this)

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
* master
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch dummy-branch
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
  dummy-branch
* master
shiwanshukumarjha@Shiv-2 GIT Assessment % git checkout master
Already on 'master'
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch -D dummy-branch
Deleted branch dummy-branch (was 639f191).
shiwanshukumarjha@Shiv-2 GIT Assessment % git branch
* master
```

Question 4: Handling Errors (Stash, Reset, Revert)

Objective : Learn how to manage mistakes and unfinished work.

* Tasks :-

1. Make changes to app.py but do NOT commit

```
# Version route
@app.route('/version')
def version():
    return "Application Version 1.0"
```

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")
```

2. Stash the changes (include untracked files)

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git stash -u
Saved working directory and index state WIP on master: 639f191 Add contact page route in feature branch
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

3. Check the stash list

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git stash list
stash@{0}: WIP on master: 639f191 Add contact page route in feature branch
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

4. Apply the stashed changes back

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git stash apply
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py

no changes added to commit (use "git add" and/or "git commit -a")
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

5. Commit the changes

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git add app.py
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app.py

shiwanshukumarjha@Shiv-2 GIT Assessment % git commit -m "Restore version route from stash"
[master 5f9649b] Restore version route from stash
 1 file changed, 5 insertions(+)
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is ahead of 'origin/master' by 4 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
shiwanshukumarjha@Shiv-2 GIT Assessment % git log --oneline
5f9649b (HEAD -> master) Restore version route from stash
639f191 Add contact page route in feature branch
5a4ae24 Add health check route to Flask application
79578d6 Add about page route to Flask application
367e75f (origin/master) Initialize Git project and add app.py with initial Python code
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

6. Make another commit with incorrect code

```
@app.route('/version')
def version()
    return "Application Version 1.0"
```

7. Undo the last commit using reset

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git log --oneline
5f9649b (HEAD -> master) Restore version route from stash
639f191 Add contact page route in feature branch
5a4ae24 Add health check route to Flask application
79578d6 Add about page route to Flask application
367e75f (origin/master) Initialize Git project and add app.py with initial Python code
shiwanshukumarjha@Shiv-2 GIT Assessment % git reset --soft HEAD~1
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app.py

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py
```

8. Make another commit

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git add app.py
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app.py

shiwanshukumarjha@Shiv-2 GIT Assessment % git commit -m "Restore version route correctly after reset"
[master 794bd36] Restore version route correctly after reset
 1 file changed, 10 insertions(+)
shiwanshukumarjha@Shiv-2 GIT Assessment % git status
On branch master
Your branch is ahead of 'origin/master' by 4 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
shiwanshukumarjha@Shiv-2 GIT Assessment %
```

9. Undo a commit using revert (create a new reversing commit)

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git log --oneline
794bd36 (HEAD -> master) Restore version route correctly after reset
639f191 Add contact page route in feature branch
5a4ae24 Add health check route to Flask application
79578d6 Add about page route to Flask application
367e75f (origin/master) Initialize Git project and add app.py with initial Python code
shiwanshukumarjha@Shiv-2 GIT Assessment % git revert 794bd36
[master a3bf938] Revert "Restore version route correctly after reset"
 1 file changed, 10 deletions(-)
```

10. Verify the commit history

```
shiwanshukumarjha@Shiv-2 GIT Assessment % git log --oneline
a3bf938 (HEAD -> master) Revert "Restore version route correctly after reset"
794bd36 Restore version route correctly after reset
639f191 Add contact page route in feature branch
5a4ae24 Add health check route to Flask application
79578d6 Add about page route to Flask application
367e75f (origin/master) Initialize Git project and add app.py with initial Python code
shiwanshukumarjha@Shiv-2 GIT Assessment %
```